1 Introduction and inequalities

Week 1 · Worksheet

Name _____ Date _____

Exercises

Show your work. Each exercise has a hint at the end: try without it first.

1. Sort the numbers Warm-up · Lesson 1

Order from smallest to largest: 0.3, -1/2, √2, -1, 1/3.

Hint: Write each as a decimal: √2 ≈ 1.414 and 1/3 ≈ 0.333.

2. Distances Practice · Lesson 1

Compute |3 - 8|, |-4 - 2|, and the distance between -1.5 and 2.

Hint: Subtract, then drop the sign.

3. Find the degree Warm-up · Lesson 2

Give the degree of: (a) $5x^3 - 2x + 1 = 0$; (b) $3x - 4 = 2x^2 - 2x^2$; (c) $x(x + 1)(x - 2) = 0$.

Hint: Simplify (b) first; in (c) count the factors of x.

4. Root or not? Practice · Lesson 2

Is $x = -1$ a root of $x^3 + 2x^2 - x - 2 = 0$? Is $x = 2$?

Hint: Substitute and compute carefully with the signs.

5. Three consecutive numbers Warm-up · Lesson 3

Three consecutive integers add up to 93. Find them.

Hint: Call them n, n + 1, n + 2.

6. A thrown ball Practice · Lesson 3

A ball's height is $h = -5t^2 + 15t$ metres after t seconds. When is it on the ground?

Hint: Set $h = 0$ and factor out $-5t$.

7. Use the formula Challenge · Lesson 3

Solve $2x^2 - 3x - 4 = 0$ and give the roots to two decimals.

Hint: $a = 2$, $b = -3$, $c = -4$, so $\Delta = 9 + 32 = 41$.

8. Factor a cubic Practice · Lesson 4

Solve $x^3 + 2x^2 - x - 2 = 0$.

Hint: Group: $x^2(x + 2) - (x + 2)$.

9. Absolute value equation Practice · Lesson 4

Solve $|3x - 1| = 8$.

Hint: $3x - 1$ is 8 or -8.

10. A fraction equation Challenge · Lesson 4

Solve $x/(x - 2) + 1 = 2/(x - 2)$.

Hint: Multiply by $(x - 2)$, then check the answer against the denominator.

11. Translate Warm-up · Lesson 5

Write as intervals: (a) $x > -1$; (b) $0 \le x < 4$; (c) $x \le 2$ or $x > 5$.

Hint: Decide bracket or parenthesis at each end.

12. Overlap Practice · Lesson 5

Find $[-3, 4) \cap (1, 7]$ and $[-3, 4) \cup (1, 7]$.

Hint: Draw both intervals on one line.

13. Solve and check Warm-up · Lesson 6

Solve $5x + 2 > 3x - 6$ and check one value.

Hint: Collect x on the left.

14. Brackets and a negative Practice · Lesson 6

Solve $2(x - 3) - 4 \geq 5(x + 1)$.

Hint: Expand both sides first.

15. Fair tickets Challenge · Lesson 6

A fair charges 8 to enter plus 1.5 per ride, or 4 per ride with free entry. For how many rides is the first option cheaper?

Hint: $8 + 1.5r < 4r$.

16. Quadratic Practice · Lesson 7

Solve $x^2 - 2x \geq 15$.

Hint: $x^2 - 2x - 15 = (x - 5)(x + 3)$.

17. Four factors Challenge · Lesson 7

Solve $x^4 - 5x^2 + 4 \geq 0$.

Hint: $(x^2 - 1)(x^2 - 4) = (x + 2)(x + 1)(x - 1)(x - 2)$.

18. A fraction Challenge · Lesson 7

Solve $(x + 4)/(x - 1) > 2$.

Hint: Subtract 2: $(x + 4 - 2x + 2)/(x - 1) = (6 - x)/(x - 1) > 0$.

19. Inside and outside Warm-up · Lesson 8

Solve $|x + 1| < 3$ and $|x + 1| \geq 3$.

Hint: The centre is -1.

20. Heights within two deviations Practice · Lesson 8

Typical heights h (cm) satisfy $|(h - 165)/8| \leq 2$. Find the range.

Hint: Multiply by 8 first: $|h - 165| \leq 16$.

21. A system with a quadratic Challenge · Lesson 8

Solve the system $2x - 4 < 6$ and $x^2 - 4x \geq 12$.

Hint: $x < 5$; and $(x - 6)(x + 2) \geq 0$ gives $x \leq -2$ or $x \geq 6$.

Quick check

Circle one answer.

22. Which set contains $\sqrt{7}$?

A) The integers $\mathbb{Z}$ B) The rationals $\mathbb{Q}$ C) The reals $\mathbb{R}$

23. What is the degree of $x(x - 1) = x^2 + 4$?

A) 2 B) 1 C) 0

24. How many real roots has $4x^2 - 4x + 1 = 0$?

A) Two B) One (repeated) C) None

25. A rectangle is 2 m longer than it is wide and has area 63 m². Solving $w(w + 2) = 63$ gives $w = 7$ or $w = -9$. What is the width?

A) 7 m B) -9 m C) Both

26. A classmate solved $x^2 = 16$ and wrote "$x = 4$". What is missing?

A) The root $x = -4$ B) Nothing C) The root $x = 8$

27. Which is a root of $x^3 - 7x + 6 = 0$?

A) $x = 3$ B) $x = 1$ C) $x = -1$

28. What is $(-\infty, 3] \cap (0, 5)$?

A) $(0, 3]$ B) $(-\infty, 5)$ C) $[0, 3)$

29. Solve $6 - 2x \geq 10$.

A) $x \geq -2$ B) $x \leq -2$ C) $x \leq 2$

Answer key

1. $-1 < -1/2 < 0.3 < 1/3 < \sqrt{2}$.
2. $|-5| = 5$; $|-6| = 6$; $|-1.5 - 2| = 3.5$.
3. (a) 3; (b) 1, because the right side is 0; (c) 3.
4. $x = -1$: $-1 + 2 + 1 - 2 = 0$, so yes. $x = 2$: $8 + 8 - 2 - 2 = 12$, so no.
5. $3n + 3 = 93$, so $n = 30$: the numbers are 30, 31, 32.
6. $-5t(t - 3) = 0$, so $t = 0$ (the throw) and $t = 3$ s (the landing).
7. $x = (3 \pm \sqrt{41})/4$, about 2.35 and −0.85.
8. $(x + 2)(x^2 - 1) = (x + 2)(x - 1)(x + 1) = 0$, so $x = -2, 1, -1$.
9. $3x - 1 = 8$ gives $x = 3$; $3x - 1 = -8$ gives $x = -7/3$.
10. $x + x - 2 = 2$ gives $x = 2$, but $x = 2$ makes the denominator zero. There is no solution.
11. (a) $(-1, \infty)$; (b) $[0, 4)$; (c) $(-\infty, 2] \cup (5, \infty)$.
12. Intersection $(1, 4)$; union $[-3, 7]$.
13. $2x > -8$, so $x > -4$: $(-4, \infty)$. With $x = 0$: $2 > -6$ is true.
14. $2x - 10 \geq 5x + 5$, so $-3x \geq 15$ and $x \leq -5$: $(-\infty, -5]$.
15. $8 < 2.5r$, so $r > 3.2$: from 4 rides on.
16. $(-\infty, -3] \cup [5, \infty)$.
17. $(-\infty, -2] \cup [-1, 1] \cup [2, \infty)$.
18. $(1, 6)$.
19. $(-4, 2)$, and $(-\infty, -4] \cup [2, \infty)$. Together they cover all of $\mathbb{R}$.
20. $149 \leq h \leq 181$, so $[149, 181]$.
21. Only $x \leq -2$ overlaps with $x < 5$: $(-\infty, -2]$.
22. **C** — $\sqrt{7}$ is irrational: it is a real number but not a fraction.
23. **B** — $x^2 - x = x^2 + 4$ simplifies to $-x - 4 = 0$: degree 1.
24. **B** — $\Delta = 16 - 16 = 0$. In fact $4x^2 - 4x + 1 = (2x - 1)^2$, so $x = 1/2$.
25. **A** — A width cannot be negative, so discard −9.
26. **A** — $(-4)^2 = 16$ as well: $x = \pm 4$.
27. **B** — $1 - 7 + 6 = 0$. (The others: $27 - 21 + 6 = 12$ and $-1 + 7 + 6 = 12$.)
28. **A** — The overlap starts just after 0 and ends at 3, included.
29. **B** — $-2x \geq 4$; dividing by −2 flips the sign: $x \leq -2$.